

WEEK 3 – POWER BI REPORT

Shopify Stock Data Analysis

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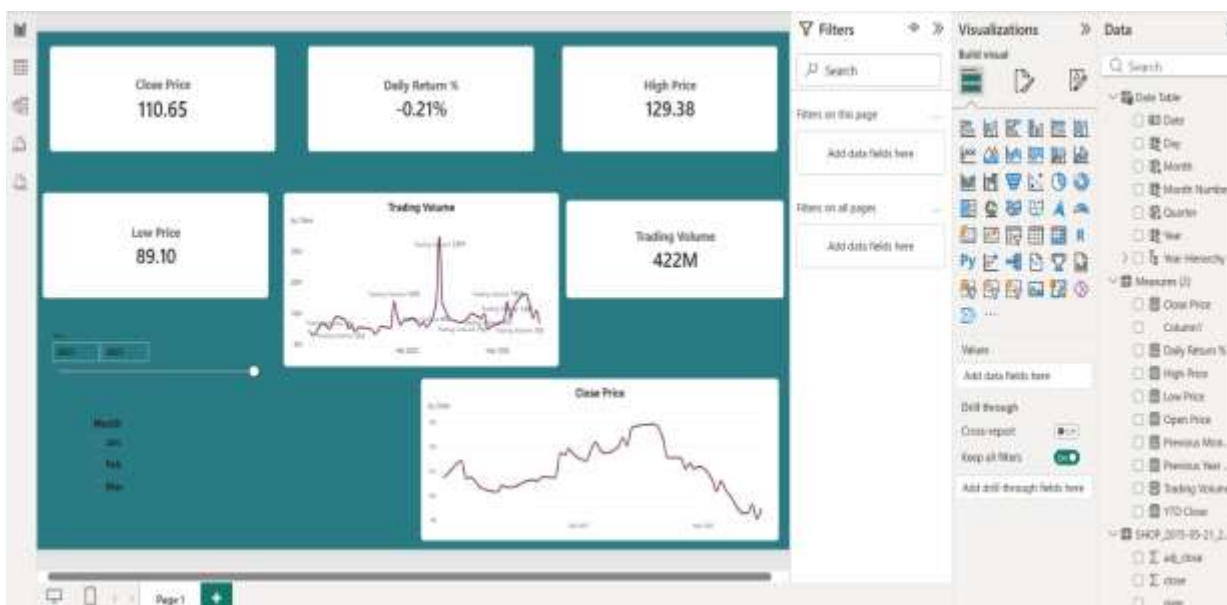
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1. Introduction

In Week 3, I worked on the Shopify stock dataset using Power BI. I created the required date structure, calculations, charts, cards and filters to build an interactive stock analysis report.

Work Completed

- Created a Date Table.
- Created a relationship between the Date Table and Shopify stock table.
- Created DAX measures for stock analysis.
- Created a Year Hierarchy.
- Created KPI cards.
- Created Close Price and Trading Volume charts.
- Created Year and Month slicers.
- Arranged and formatted the final dashboard.



2. Importing and Understanding the Data

I first loaded the Shopify stock dataset into Power BI and checked the available columns.

The main columns used in the analysis were Date, Open, High, Low, Close, Adjusted Close and Volume.

The Date column was important because the report needed to analyse stock information over different time periods.

	A	B	C	D	E	F	G	H
	date	open	high	low	close	adj_close	volume	
	2015-05-2	2.8	2.874	2.411	2.568	2.568	1.23E+08	
	2015-05-2	2.607	3.11	2.6	2.831	2.831	28412000	
	2015-05-2	2.98	3.034	2.908	2.965	2.965	8202000	
	2015-05-2	3.067	3.081	2.7	2.75	2.75	7976000	
	2015-05-2	2.755	2.774	2.648	2.745	2.745	4000000	
	2015-05-2	2.725	2.738	2.687	2.72	2.72	2554000	
	2015-06-0	2.72	2.744	2.66	2.733	2.733	2569000	
	2015-06-0	2.747	2.747	2.665	2.695	2.695	3798000	
	2015-06-0	2.676	2.75	2.65	2.744	2.744	3024000	
	2015-06-0	2.742	2.749	2.551	2.678	2.678	3226000	
	2015-06-0	2.666	2.739	2.61	2.739	2.739	1294000	
	2015-06-0	2.69	2.748	2.682	2.698	2.698	2809000	
	2015-06-0	2.664	2.846	2.664	2.834	2.834	3375000	
	2015-06-1	2.845	2.888	2.795	2.81	2.81	2734000	
	2015-06-1	2.849	3.073	2.849	3.073	3.073	4606000	
	2015-06-1	3.132	3.37	3.125	3.363	3.363	8474000	
	2015-06-1	3.397	3.812	3.292	3.55	3.55	16943000	
	2015-06-1	3.606	3.8379	3.584	3.811	3.811	11095000	
	2015-06-1	3.963	4.213	3.801	3.863	3.863	15952000	
	2015-06-1	3.842	3.889	3.4326	3.555	3.555	11482000	
	2015-06-1	3.633	3.674	3.255	3.552	3.552	9797000	
	2015-06-2	3.676	3.85	3.609	3.694	3.694	7922000	
	2015-06-2	3.775	3.81	3.517	3.543	3.543	2939000	
	2015-06-2	3.512	3.7429	3.414	3.596	3.596	5644000	
	2015-06-2	3.634	3.65	3.48	3.484	3.484	2082000	
	2015-06-2	3.487	3.532	3.263	3.352	3.352	3465000	
	2015-06-3	3.373	3.373	3.43	3.433	3.433	3455000	

SHOP 2015-05-21 2025-03-16

3. Creating the Date Table

I created a separate Date Table to organise the date information used in the report.

The Date Table contains Date, Day, Month, Month Number, Quarter and Year.

I then marked the Date Table as a Date Table in Power BI by selecting the Date column.

1 Date Table = CALENDAR(MIN('SHOP_2015-05-21_2025-03-16'[date]),MAX('SHOP_2015-05-21_2025-03-16'[date]))

Date	Day	Month	Month Number	Quarter	Year
21-05-2015 00:00:00	21	May		5 Q2	2015
22-05-2015 00:00:00	22	May		5 Q2	2015
23-05-2015 00:00:00	23	May		5 Q2	2015
24-05-2015 00:00:00	24	May		5 Q2	2015
25-05-2015 00:00:00	25	May		5 Q2	2015
26-05-2015 00:00:00	26	May		5 Q2	2015
27-05-2015 00:00:00	27	May		5 Q2	2015
28-05-2015 00:00:00	28	May		5 Q2	2015
29-05-2015 00:00:00	29	May		5 Q2	2015
30-05-2015 00:00:00	30	May		5 Q2	2015
31-05-2015 00:00:00	31	May		5 Q2	2015
01-06-2015 00:00:00	1	Jun		6 Q2	2015
02-06-2015 00:00:00	2	Jun		6 Q2	2015
03-06-2015 00:00:00	3	Jun		6 Q2	2015
04-06-2015 00:00:00	4	Jun		6 Q2	2015
05-06-2015 00:00:00	5	Jun		6 Q2	2015
06-06-2015 00:00:00	6	Jun		6 Q2	2015
07-06-2015 00:00:00	7	Jun		6 Q2	2015
08-06-2015 00:00:00	8	Jun		6 Q2	2015
09-06-2015 00:00:00	9	Jun		6 Q2	2015
10-06-2015 00:00:00	10	Jun		6 Q2	2015
11-06-2015 00:00:00	11	Jun		6 Q2	2015
12-06-2015 00:00:00	12	Jun		6 Q2	2015
13-06-2015 00:00:00	13	Jun		6 Q2	2015
14-06-2015 00:00:00	14	Jun		6 Q2	2015
15-06-2015 00:00:00	15	Jun		6 Q2	2015
16-06-2015 00:00:00	16	Jun		6 Q2	2015
17-06-2015 00:00:00	17	Jun		6 Q2	2015

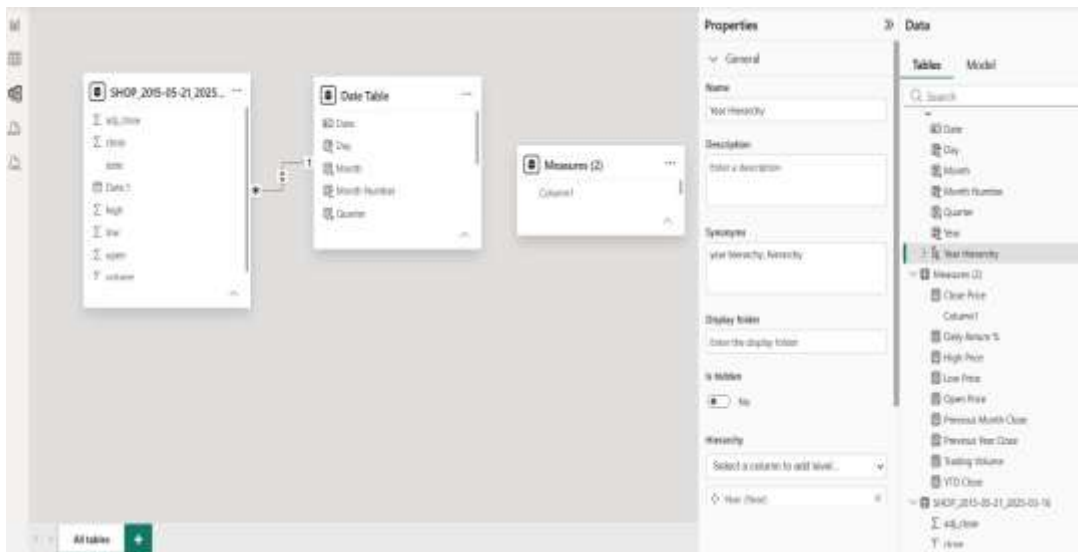
Table: Date Table (3,586 rows)

4. Creating the Relationship

After creating the Date Table, I connected it with the Shopify stock table using the Date column.

The relationship is a one-to-many relationship. The Date Table contains the date values, and the stock table contains the related stock records.

This relationship allows the date fields and slicers to work with the stock data correctly.

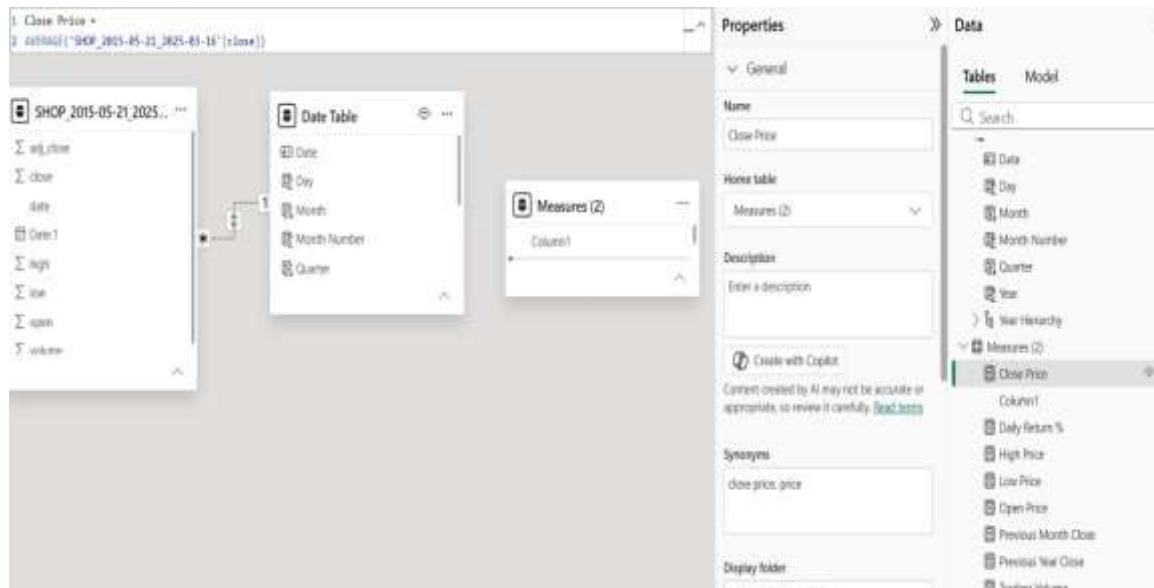


5. Creating DAX Measures

I created DAX measures to calculate the important stock values used in the report.

- Open Price – average opening price.
Open Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[open])
- Close Price – average closing price.
Close Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[close])
- High Price – highest stock price.
High Price = MAX('SHOP_2015-05-21_2025-03-16'[high])
- Low Price – lowest stock price.
Low Price = MIN('SHOP_2015-05-21_2025-03-16'[low])
- Trading Volume – total trading volume.
- Daily Return % – return calculated from the Open Price and Close Price.
- Previous Month Close – previous month's closing value.
- YTD Close – year-to-date closing value.
YTD Close = TOTALYTD([Close Price], 'Date Table'[Date])
- Previous Year Close – previous year's closing value.
Previous Year Close = CALCULATE([Close Price],
SAMEPERIODLASTYEAR('DateTable'[Date]))

These measures were used in the cards and other report visuals.

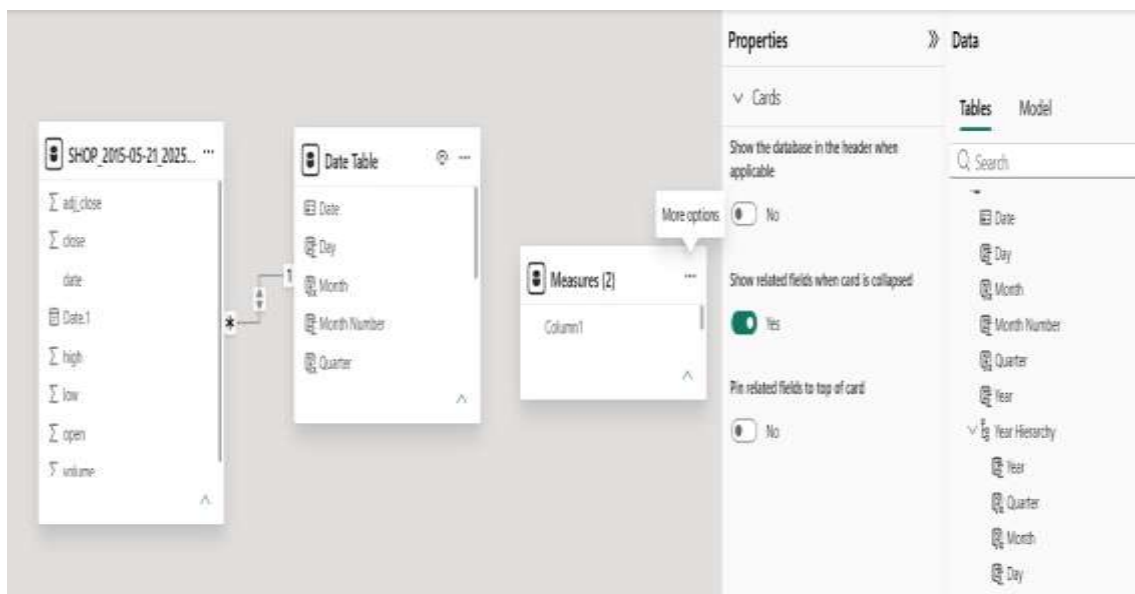


6.Creating the Year Hierarchy

I created a Year Hierarchy from the Date Table for time-based analysis.

The hierarchy contains four levels: Year → Quarter → Month → Day.

I also sorted the Month column using Month Number so that the months appear in the correct order from January to December.



7. Creating KPI Cards

I created KPI cards to show the important stock values clearly on the dashboard.

- Close Price
- Daily Return %
- High Price
- Low Price
- Trading Volume

The cards provide a quick view of the main values without having to check the complete dataset.



8. Creating Charts and Slicers

Charts

I created a Close Price line chart to show the change in Shopify's closing price over time.

I also created a Trading Volume line chart to show trading volume over different dates.

Slicers

I created Year and Month slicers. These slicers allow the report to be filtered by a selected year or month.



9. Final Dashboard and Conclusion

Final Dashboard

Finally, I arranged the cards, charts and slicers together on one Power BI report page. I also formatted the visuals so that the information is clear and easy to read.

- Close Price card
- Daily Return % card
- High Price card
- Low Price card
- Trading Volume card
- Close Price trend chart
- Trading Volume trend chart
- Year slicer
- Month slicer

Conclusion

In Week 3, I completed the Shopify stock analysis report in Power BI. I created the Date Table, relationship, DAX measures, hierarchy, KPI cards, charts and slicers. The final dashboard provides a simple interactive view of Shopify stock prices and trading volume.

